Brahmastra: A Post-Quantum Self-Sovereign Identity Protocol with Cryptographic Tar-Pit Defense and Autonomous Dharma Economics

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***Abstract*—We present Brahmastra, a decentralized post-quantum zero-knowledge identity network designed to mitigate pervasive surveillance. The protocol integrates Sphinx onion routing, lattice-based cryptography, algorithmic tar-pits, and Dharma Economics for autonomous welfare routing.**

***Index Terms*—Post-Quantum Cryptography, Zero-Knowledge Proofs, Tar-Pit Defense, Dharma Economics.**

I. Introduction

Modern distributed networks suffer from metadata correlation vulnerabilities. Brahmastra mitigates unauthorized traffic analysis via post-quantum cryptography and delay-based cryptographic tar-pits.

II. Cryptographic Architecture and Tar-Pit Defense

Relay nodes enforce deterministic delay responses against probing adversaries:

$$T_{\text{delay}} = \lambda \cdot 2^{\kappa} + \mathcal{H}(\text{sender_pk}) \tag{1}$$

where κ denotes probe intensity and $\mathcal{H}$ enforces non-linear computational state expansion.

III. Dharma Economics Treasury Model

Protocol network settlements enforce an immutable treasury split:

$$\mathcal{T}_{\text{inflow}} \rightarrow 0.80 \cdot \mathcal{W}_{\text{welfare}} + 0.20 \cdot \mathcal{I}_{\text{infra}} \tag{2}$$

Disbursements are executed via zero-knowledge direct benefit transfers to agricultural and healthcare beneficiaries.

IV. Conclusion

Brahmastra provides a resilient, mathematically verified infrastructure aligned with Swiss foundation governance standards.